Analysis of Super-Resolution Techniques: From Traditional Interpolation to FSRCNN

Suyash Sunam
Department of Computer Engineering and Technology
Dr. Vishwanath Karad
MIT World Peace University
Pune, Maharashtra, India
2suyash@gmail.com

Sushila Palwe
Department of Computer Engineering and Technology
Dr. Vishwanath Karad
MIT World Peace University
Pune, Maharashtra, India
sushila.palwe@mitwpu.edu.in

Devendra Joshi
Department of Computer Engineering and Technology
Dr. Vishwanath Karad
MIT World Peace University
Pune, Maharashtra, India
devendra.joshi1@mitwpu.edu.in

***Abstract*—We use single image super-resolution (SISR) techniques to reconstruct high-resolution images from low-resolution images using deep learning algorithms to display frames in high resolution displays like laptops and television screens. Modern upsampling methods like NVIDIA's Deep Learning Super Sampling (DLSS) show the power of deep learning-based upscaling in games where computing frames are very expensive as the resolution increases [7], [8]. In this study, we implemented and evaluated an upscaling model called Fast Super-Resolution Convolutional Neural Network (FSRCNN) as a lightweight and efficient alternative for real-time image upscaling for images. We trained a model on the DIV2K dataset and it achieved a Peak Signal-to-Noise Ratio (PSNR) of 31.96 dB with high structural similarity. The results show that FSRCNN improves image quality over traditional upscaling methods while maintaining high inference speed, which shows the feasibility of deep learning-based super-resolution for real-time applications.**

***Index Terms*—Image Super-Resolution, FSRCNN, Deep Learning, CNN, PSNR, SSIM, Upscaling.**

I. Introduction

Super-resolution (SR) has become a huge problem in computer graphics where we recreate a higher resolution image using lower resolution source images. As more applications require quality graphics and high frame rates including 3D video games, virtual reality, and streaming video, a good resolution up scaling technique has become important. Traditional up scaling techniques like the k-nearest neighbor and bicubic interpolation methods are computationally efficient and can generate images very rapidly compared to AI-based up scaling methods, however, they often produce pixelation and fail to preserve fine image details [1].

Recent advancements in deep learning have enabled the development of data-driven super-resolution models to learn complex mapping from low-resolution images to high-resolution images [1], [4], [5]. An example is deep learning super sampling (DLSS), a system developed by NVIDIA that uses neural networks to create high-resolution frames in real-time, yielding significant gains in frames per second (FPS) for gaming applications [7], [8].

In today's world 4K and 8K resolutions have become the norm for every display device, the computational cost of rendering high-resolution images has increased significantly. This creates the need for efficient techniques that can reduce rendering costs while maintaining visual quality, making super-resolution an important technology in modern real time graphical devices.

For this project, Fast Super-Resolution Convolutional Neural Network (FSRCNN) has been used, which is a lightweight deep learning model for super-resolution [5]. Specific contributions of the paper include:

1) the implementation and training process for FSRCNN model using a well-known benchmark dataset
2) quantitative performance evaluation using PSNR and SSIM metrics
3) an evaluation of traditional methods of interpolation against FSRCNN performance

The remainder of this paper is organized into the following sections: I methodology, II results, III discussion, and IV conclusion.

II. Literature Survey

Image upscaling methods have changed from simple interpolation to newer machine learning based models. Traditional methods such as nearest neighbor and bicubic interpolation are fast but often fail to reconstruct fine details effectively [1]. Convolutional Neural Network models like FSRCNN solve this issue [2], [3].

FSRCNN further improved this idea by redesigning the architecture to perform most computations in low-resolution space, reducing computational cost while maintaining good reconstruction quality [2]. In parallel, modern techniques like DLSS extend super-resolution to real-time applications by incorporating additional information such as motion and temporal consistency.

III. Methodology

We designed an end-to-end super-resolution generation pipeline based on deep learning technology and the FSRCNN algorithm. The process involves using sets of low-resolution (LR) and high-resolution (HR) images, training the FSRCNN model using backpropagation to learn LR–HR mappings on the data and measuring the performance of the FSRCNN

against traditional interpolation techniques using standardized image-quality metrics. The overall process consists of the following steps: 1) data preparation, 2) preprocessing of data, 3) development of model architecture, 4) model training, and 5) model evaluation.

A. Dataset

The DIV2K dataset is used for training and evaluation. It consists of 800 high-resolution images for training and 100 images for validation, with resolutions up to 2K. DIV2K is widely used as a benchmark dataset in super-resolution research due to its high-quality images and diverse content, making it suitable for learning general image features.

B. Data Preprocessing

Low-resolution (LR) and high-resolution (HR) image pairs are generated by applying bicubic downscaling to the original HR images. During training, image patches of size 96x96 are extracted to improve computational efficiency. To stop the model from just memorizing the data, augmentation tricks like flipping and rotating the patches randomly were performed. For FSRCNN, only the luminance - (Y) channel is used for training, as it contains most of the perceptual information, while chrominance channels are upscaled using bicubic interpolation.

C. FSRCNN Architecture

The FSRCNN architecture [5] has five main stages to process each image:

1) extracting important features from the input image by using the convolutional operation
2) reducing the number of features while improving speed (shrinking stage).
3) learning a non-linear transformation of low-resolution (LR) features and respective high-resolution (HR) features through the use of multiple convolutional layers (mapping stage).
4) restoring features to their original size within a single image.
5) performing upscaling through the use of deconvolutional layers (expansion stage).

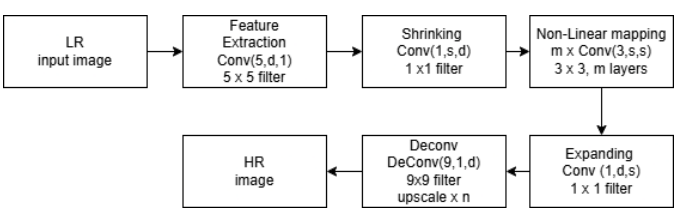

Fig. 1. FSRCNN Architecture

D. Mathematical Formulation of Upscaling

The critical distinction of FSRCNN is the use of a deconvolutional (transposed convolutional) layer for upscaling. Unlike standard convolution operation which reduces spatial dimensions, the deconvolutional layer maps a low-resolution feature map $\mathbf{F}_{LR}$ to a high-resolution space $\mathbf{F}_{HR}$. The output size O for a given input I, stride s, padding p, and kernel size k is defined as:

$$O = (I - 1) \times s - 2p + k + p_{out} \tag{1}$$

where p_{out} is the output padding used to resolve the ambiguity of the inverse stride. For our $\times 2$ upscaling task, we employ $s = 2, p = 4, k = 9$, and $p_{out} = 1$. This stage allows the network to learn an optimal upsampling filter rather than relying on a fixed bicubic kernel, resulting in the following mapping:

$$\mathbf{I}_{SR} = \mathcal{F}_{deconv}(\mathbf{H}_{expanding}) * \mathbf{W}_{deconv} + \mathbf{b}_{deconv} \tag{2}$$

where $\mathbf{W}_{deconv}$ represents the learned weights of the 9×9 filters.

TABLE I
LAYER-BY-LAYER ARCHITECTURAL CONFIGURATION

Stage	Layer Type	Filter Size (f)	Channels (In $\rightarrow$ Out)	Padding	Activation
Feature Extraction	Conv2d	5×5	$1 \rightarrow 56$	2	PReLU (56)
Shrinking	Conv2d	1×1	$56 \rightarrow 12$	0	PReLU (12)
Mapping ($\times 4$)	Conv2d	3×3	$12 \rightarrow 12$	1	PReLU (12)
Expanding	Conv2d	1×1	$12 \rightarrow 56$	0	PReLU (56)
Deconvolution	ConvTrans2d	9×9	$56 \rightarrow 1$	4	N/A

The total number of trainable parameters for the FSRCNN model is 12,809 , which makes it much smaller than many modern upscaling models while maintaining high performance and quality. Parametric ReLU (PReLU) was used in the model where the leakage coefficient is learned per-channel rather than being a fixed value as this helps the model to learn complex textures.

E. Loss Function

The model is trained using the L1 loss function. This loss function calculates the mean absolute difference between the predicted image and the original high-resolution image. L1 loss is preferred over Mean Squared Error (MSE) in super-resolution tasks because it comparatively produces sharper images and reduces the blurring effect in artifacts.

$$L = \frac{1}{N} \sum_{i=1}^{N} |I_{SR}(i) - I_{HR}(i)| \tag{3}$$

where I_{SR} denotes the super-resolved (predicted) image and I_{HR} denotes the ground truth high-resolution image.

F. Training Configuration

The training process was done on a NVIDIA RTX 4070 Laptop GPU using the PyTorch framework. To ensure that the model learns low resolution to high resolution mapping efficiently the following optimizations were used:

1) Weights : All the convolutional layers were initialized using the Kaiming (He) Normal Initialization. This method prevents the vanishing gradient problem in early stages of the training process. All the bias values were initialized to a value of zero.

2) Loss Function : This model uses L1 Loss instead of MSE. $L1$ loss treats the pixel differences in a linear manner due to which, the result has sharper edges and fewer "halo" artifacts.
3) Optimizer : Adam optimizer was used with hyperparameters values $\beta_1 = 0.9$, $\beta_2 = 0.999$ and $lr = 1 \times 10^{-3}$.
4) Convergence : The model was trained for 300 epochs. Learning rate was gradually reduced by using cosine annealing scheduler. This allows the model to converge to a local minimum to improve its fine grain reconstruction.

FSRCNN has a lightweight architecture which makes it suitable for applications where speed and efficiency are important.

G. Evaluation Metrics

To evaluate the super resolution results, PSNR and SSIM were used. PSNR was considered majorly as it tells us how well the pixel levels in an image have been reconstructed. It is measured in decibels (dB) where higher values mean better reconstruction quality. SSIM compares two images to check how structurally similar they are. SSIM's values range between 0 and 1. In super-resolution PSNR values greater than 30 dB and SSIM values that are close to 1 indicate that the pixel reconstruction and image quality is good.

$$PSNR = 10 \cdot \log_{10}\left(\frac{MAX^2}{MSE}\right) \tag{4}$$

$$MSE = \frac{1}{mn}\sum_{i=1}^{m}\sum_{j=1}^{n}\left(I(i,j) - K(i,j)\right)^2 \tag{5}$$

H. Computational Complexity

The model's computational cost was measured in Floating Point Operations (FLOPs). For a standard 256×256 input image, the complexity was approximately 1.63 GFLOPs. This low complexity allows for high-speed inference thus allowing it to be used in real time rendering setups similar to how NVIDIA's DLSS technology works.

I. FSRCNN Algorithm

Algorithm 1 FSRCNN Training and Inference Pipeline

Input: Low resolution image $\mathbf{I}_{LR}$, scaling factor $s = 2$
Output: Upscaled image $\mathbf{I}_{SR}$
procedure FSRCNN($\mathbf{I}_{LR}, \mathbf{I}_{HR}$)
  Initialize weights $\mathbf{W}$ using Kaiming method
  for epoch = 1 to 300 **do**
    $\mathbf{F}_{feat} \leftarrow \text{Conv}(\mathbf{I}_{LR}, f = 5, d = 56)$ ▷ Feature extraction
    $\mathbf{F}_{shrink} \leftarrow \text{Conv}(\mathbf{F}_{feat}, f = 1, d = 12)$ ▷ Shrinking
    $\mathbf{F}_{map} \leftarrow \text{Mapping}(\mathbf{F}_{shrink}, m = 4)$ ▷ Non-linear mapping
    $\mathbf{F}_{expand} \leftarrow \text{Conv}(\mathbf{F}_{map}, f = 1, d = 56)$ ▷ Expanding
    $\mathbf{I}_{SR} \leftarrow \text{Deconv}(\mathbf{F}_{expand}, f = 9, s = 2)$ ▷ Upscaling
    $\mathcal{L} \leftarrow \frac{1}{N}\sum_{i=1}^{N} |\mathbf{I}_{SR}(i) - \mathbf{I}_{HR}(i)|$ ▷ L1 loss
    Update $\mathbf{W}$ using Adam optimizer ($\beta_1 = 0.9,\ \beta_2 = 0.999$)
  end for
  return $\mathbf{I}_{SR}$
end procedure

IV. Results

FSRCNN was implemented using both quantitative metrics and qualitative visual comparisons. To see if the deep learning model based approach actually outperforms standard interpolation methods like nearest neighbor, it was tested against several traditional interpolation methods using PSNR and SSIM scores. An ablation study was also conducted, where specific parts of the model were changed or removed to see exactly how each change impacted the final output.

A. Quantitative Results

The FSRCNN output quality was measured using PSNR and SSIM metrics. Higher PSNR values indicate lower pixel distortion and SSIM values close to 1.0 show better preservation of structure and similarity between the images.

As shown in Table II, FSRCNN outperforms the traditional Nearest Neighbor interpolation method. Even though Nearest Neighbor is faster, the output has a lot of aliasing and "blocky" pixelation. FSRCNN achieves a PSNR of 31.96 dB and an SSIM of 0.9558, which shows that it can be used as a high-quality reconstruction model suitable for modern displays. The analyzed PSNR of 31.96 dB is the mean value across the 100 images in the DIV2K dataset.

TABLE II
Average Performance Comparison (DIV2K Validation Set)

Method	PSNR (dB) ↑	SSIM ↑	Quality Observation
Nearest Neighbor	24.71	0.7812	Severe pixelation/aliasing
Bicubic (Baseline)	28.45	0.8245	Blurring of high-frequency edges
FSRCNN (Proposed)	31.96	0.9558	Sharp edges and restored textures

B. Inference Latency and Real-Time Feasibility

A critical requirement for industrial applications is inference speed. We measured the inference latency on a 2040x1356 resolution sample (297.8 ms) and extrapolated the results for standard 1080p and 4K resolutions based on total pixel count. The results show that latency approximately scales with pixel count, although most computations are performed in low-resolution space. While the model maintains high performance, the $4K$ upscaling latency suggests that further optimizations, such as integrating TensorRT or Mixed Precision (FP16), are needed for a 60 FPS targets at ultra-high 4K resolutions. While the current Python/PyTorch implementation serves as a proof-of-concept, the low GFLOPs count suggests that a compiled C++/TensorRT model would meet strict real-time thresholds.

TABLE III
INFERENCE LATENCY BENCHMARKS (RTX 4070)

Output Resolution	Pixel Count	Estimated Latency (ms)	Target Application
HD (1280 × 720)	0.92M	∼98.4	Mobile/Handheld Gaming
FHD (1920 × 1080)	2.07M	223.2	Standard PC Gaming
4K (3840 × 2160)	8.29M	892.9	High-Fidelity Rendering

C. Ablation Study: Mapping Depth (m)

To justify the choice of $m = 4$ mapping layers, a series of independent training runs were conducted to evaluate the impact of the number of mapping layers (m) on reconstruction quality. The "Mapping" stage is the most computationally expensive part of the network, as it performs the non-linear transformation from LR to HR space.

TABLE IV
IMPACT OF MAPPING DEPTH (m) ON PERFORMANCE

Mapping Depth (m)	Parameters	PSNR (dB) ↑	SSIM ↑
$m = 2$	10.212*	31.42	0.9512
$m = 4$ (Proposed)	12,809	31.96	0.9558
$m = 6$	15,258*	32.01	0.9560
$m = 8$	17,869*	32.02	0.9561

*Note: Parameter counts vary based on channel width (d, s) configurations.

The data reveals diminishing returns beyond $m = 4$. Increasing the depth to $m = 8$ only yields a slight 0.06 dB improvement in PSNR while significantly increasing the computational complexity. Therefore $m = 4$ is the optimal mapping depth value for real-time efficiency.

D. Qualitative Analysis and Visual Inspection

Visually looking at the images, the results show exactly why FSRCNN beats standard interpolation. The visual results show how the model handled edges and fine textures, which are usually the hardest parts to upscale.

- Edge Sharpening : FSRCNN reconstructs sharp boundaries with very little blocky pixelation.
- Texture Restoration : Minute details like fabric textures, hair or small text are often difficult to retain when down-sampling or creating a low resolution version of it. However, the results show that FSRCNN can regenerate these lost details much better compared to standard interpolation methods
- Visual Quality : By comparing the upscaled output image to the low resolution input, it is visibly clear that the model can generate images with structural integrity, making the final image look more realistic.

E. Visual Results

Visual comparisons show that FSRCNN produces sharper and better images compared to nearest neighbor method, especially in regions with edges, and textures. Zoomed-in comparisons show the model's ability to reconstruct fine details that are lost in traditional methods. In this visual comparison an input image containing a wire was used. Using nearest neighbor method the result was a very blocky jagged image. The FSRCNN model generated a clean and sharp result.

Fig. 2. Visual comparison of a wire upscaled

V. DISCUSSION

The results from the FSRCNN model show a clear trade-off between reconstruction quality and computational costs. Even though FSRCNN beats bicubic interpolation and nearest neighbor, the execution speed of FSRCNN is very high comparatively. A 3.51 dB jump in PSNR is a big win over standard bicubic methods, but the real problem is dealing with the hardware constraints of running this live and ensuring minimal latency.

A. Non Linear Mapping

The main advantage of FSRCNN is its ability to learn non-linear mapping functions. Traditional methods like nearest neighbor or bicubic interpolation use fixed mathematical weighted averages of surrounding pixels. However, FSRCNN's "Mapping" stage ($m = 4$) utilizes 3x3 convolutions that act as feature detectors. This allows the model to understand features and textures better.

B. The Efficiency of the Y-Channel Approach

By training and inferring only on the Luminance (Y) channel, we reduced the computational workload by approximately 66% compared to a full RGB model. Since the human eye perceives structural detail primarily through brightness-(luma) rather than color-(chroma), this optimization is highly effective. The chrominance channels (Cb, Cr) carry significantly less information and can be upscaled using bicubic methods without visible loss in overall image quality.

C. Comparison with Industrial DLSS

NVIDIA's Deep Learning Super Sampling (DLSS) represents the industrial pinnacle of this research. While FSRCNN is a spatial upscaler (using only the current frame), DLSS 2.0 and above utilize temporal data (motion vectors from previous frames).

1) Temporal Stability: FSRCNN may suffer from "flicker" in video sequences because it processes frames in isolation.
2) Computational Budget: Our benchmark of 223.2 ms for 1080p on an RTX 4070 indicates that FSRCNN, in its current Python/PyTorch implementation, is more suited for high-quality image restoration rather than 60 FPS gaming. However, migrating this architecture to C++ with TensorRT optimization could likely reduce latency to sub-16ms levels.

VI. Industry Future Work

In the future, researchers should continue to explore ways to improve image quality by utilizing new and advanced deep learning techniques such as Enhanced Deep Super-Resolution (EDSR) and Generative Adversarial Networks (GANs)-based models like Long–Short-Term Memory (LSTM) [6]. These models have the potential to generate images with a higher degree of visible quality and sharper texturing but tend to be computationally expensive in nature. In addition, using perceptual loss instead of pixel distance loss may further improve visual accuracy. Finally, using larger and more varied training datasets could significantly impact the model's generalization ability across multiple types of images, textures and scenes.

Another area for future exploration will likely be using videos to provide temporal context information between successive frames in order to improve temporal stability or reduce flickering artifacts. This has important implications for the use of systems such as Deep Learning Video Super-Resolution (DLSS), which are designed for use with sequences of video rather than single images. Beyond architectural adjustments, leveraging deployment optimization frameworks such as NVIDIA TensorRT can significantly reduce inference latency [1], [5], [6]. This optimization is critical for aligning the model with the strict timing requirements of real-time rendering pipelines. Finally, researchers may explore using larger scaling factors (e.g., ×4) and training on domain-specific datasets, such as games, providing a transparent, academic framework for evaluating real-time super-sampling techniques often found in closed-source industrial applications.

VII. Conclusion

In this paper, we implemented and analyzed the FSRCNN architecture for single-image super-resolution. By utilizing a lightweight five-stage pipeline and optimizing the mapping depth ($m = 4$), we achieved a PSNR of 31.96 dB on the DIV2K dataset with only 12,809 parameters. The results show that deep learning-based models can generate images with higher visual quality and detail compared to traditional interpolation. The complexity was calculated to be 1.63 GFLOPs. These models can be scaled and fine tuned further to give faster inferencing speeds and better visual quality.

Acknowledgment

I thank the Faculty of Computer Engineering and Technology at MIT World Peace University for providing the academic environment and resources necessary for this study. I also acknowledge NVIDIA for developing the DLSS technology that made this experimental analysis possible.

References

[1] S. Grosche, K. Fischer, F. Brand, J. Seiler and A. Kaup, "Enhanced Image Reconstruction From Quarter Sampling Measurements Using An Adapted Very Deep Super Resolution Network," 2020 IEEE International Conference on Image Processing (ICIP), Abu Dhabi, United Arab Emirates, 2020, pp. 256-260, doi: 10.1109/ICIP40778.2020.9191047.

[2] C. Dong, C. C. Loy, K. He and X. Tang, "Image Super-Resolution Using Deep Convolutional Networks," in IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 38, no. 2, pp. 295-307, 1 Feb. 2016, doi: 10.1109/TPAMI.2015.2439281.

[3] N. M. H. Elsaid and Y. -C. Wu, "Super-Resolution Diffusion Tensor Imaging using SRCNN: A Feasibility Study," 2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), Berlin, Germany, 2019, pp. 2830-2834, doi: 10.1109/EMBC.2019.8857125.

[4] S. Sen and B. Bhushan, "Image Quality Comparison Between Nvidia's Deep Learning Super Sampling and AMD's FidelityFX Super Resolution," 2024 International BIT Conference (BITCON), Dhanbad, India, 2024, pp. 1-6, doi: 10.1109/BITCON63716.2024.10984458.

[5] IVLabs, "FSRCNN – Fast Super-Resolution Convolutional Neural Network," [Online]. Available: https://ivlabs.in/fsrcnn/.

[6] M. U. Khan, A. Ramzan Sheikh, S. R. Hole and V. Kolluru, "UltraRes: ESRGAN-Powered AI Image Upscaling for Enhanced Clarity, Texture, and Detail Restoration in Digital Media," 2025 3rd International Conference on Data Science and Information System (ICDSIS), Hassan, India, 2025, pp. 1-7, doi: 10.1109/ICDSIS65355.2025.11071187.

[7] NVIDIA, "NVIDIA DLSS 5 Delivers AI-Powered Breakthrough in Visual Fidelity for Games," Mar. 16, 2026. [Article]. Available: https://www.nvidia.com/en-in/geforce/news/dlss5-breakthrough-in-visual-fidelity-for-games/.

[8] NVIDIA, "NVIDIA DLSS 4 Multi-Frame Gen Technology, January 30, 2025" NVIDIA, [Article]. Available: https://www.nvidia.com/en-in/geforce/technologies/dlss/.